

concave function of the birth probabilities; this is also pointed out in Marshall and Olkin ([11], p. 305). As a consequence of Proposition E.11.d on page 306 in that book the same holds for any c . We will give a direct elementary proof of it.

Theorem 1. *The function $P(N > n)$ is Schur-concave in $\mathbf{p}$.*

Proof. Set $f(\mathbf{p}) = P(N > n)$. The assertion is proved if we can show that $\mathbf{p}' < \mathbf{p}$ implies $f(\mathbf{p}) \leq f(\mathbf{p}')$. By symmetry it suffices to consider the case when $\mathbf{p}'$ can be obtained from $\mathbf{p}$ by a T -transformation with $j = 1$ and $k = 2$.

Let $(X_1, \dots, X_d)$ be $\text{Mult}(n, \mathbf{p})$, that is, for nonnegative integers $x_1, \dots, x_d$ with sum n , we have

$$P(X_1 = x_1, \dots, X_d = x_d) = \frac{n!}{x_1! \cdots x_d!} p_1^{x_1} \cdots p_d^{x_d}.$$

The event $N > n$ is the same as saying that in the first n trials of the experiment all outcomes have occurred less than c times. As the number of times the different outcomes have occurred has the above multinomial distribution, we have

$$P(N > n) = P(X_1 < c \cap \cdots \cap X_d < c).$$

Conditioned on $X_1 + X_2 = m$ the random vectors (X_1, X_2) and $(X_3, \dots, X_d)$ are independent and furthermore

$$\begin{aligned} \mathcal{L}(X_3, \dots, X_d | X_1 + X_2 = m) \\ &= \text{Mult}(n - m, p_3/(1 - p_1 - p_2), \dots, p_d/(1 - p_1 - p_2)), \\ \mathcal{L}(X_1 + X_2) &= \text{Bin}(n, p_1 + p_2), \\ \mathcal{L}(X_1 | X_1 + X_2 = m) &= \text{Bin}(m, p_1/(p_1 + p_2)). \end{aligned}$$

Using this, we get

$$\begin{aligned} P(N > n) &= \sum_{m=0}^n P(X_1 + X_2 = m) \cdot P(X_3 < c \cap \cdots \cap X_d < c | X_1 + X_2 = m) \\ &\quad \cdot P(m - c < X_1 < c | X_1 + X_2 = m). \end{aligned}$$

The first two factors in this sum depend only on the p 's through $p_1 + p_2$, $p_3, \dots, p_d$, and they are therefore unchanged under the T -transformation, that is, if p_1 and p_2 are replaced by $\lambda p_1 + (1 - \lambda)p_2$ and $\lambda p_2 + (1 - \lambda)p_1$, respectively.

The third factor can be written

$$g(p) = P(m - c < X_1 < c | X_1 + X_2 = m) = \sum_{k=m-c+1}^{c-1} \binom{m}{k} p^k (1-p)^{m-k},$$

where $p = p_1/(p_1 + p_2)$. In Lemma 1 below it is proved that g is increasing in the interval $(0, 1/2)$. As g is symmetric around $p = 1/2$, it follows that it increases under the T -transformation.

Combining the above conclusions the assertion follows. $\square$